Cisco ASA Firewall Complete Lab Guide (Packet Tracer)

This document contains the complete working Cisco ASA firewall lab setup created in Packet Tracer. It includes topology, IP addressing, ASA configuration, NAT, default routes, ICMP inspection, testing procedures, troubleshooting, and enterprise firewall concepts.

1. Final Topology

```
Router -------- ASA Firewall -------- PC
         outside             inside
```

2. IP Addressing Table

```
Router Fa0/0          : 192.168.0.1 /24
ASA VLAN2 (outside)  : 192.168.0.3 /24
ASA VLAN10 (inside)  : 10.0.0.1 /24
PC FastEthernet0     : 10.0.0.5 /24
Default Gateway PC   : 10.0.0.1
```

3. Physical Connections

```
Router Fa0/0  → ASA Ethernet0/0
ASA Ethernet0/1 → PC

Cable Type:
Copper Straight-Through
```

4. Router Configuration

```
enable
configure terminal

interface fastethernet0/0
ip address 192.168.0.1 255.255.255.0
no shutdown

end
```

5. ASA Configuration - Outside Interface

```
enable
configure terminal

interface vlan 2
nameif outside
security-level 0
ip address 192.168.0.3 255.255.255.0
no shutdown
```

6. Assign Ethernet0/0 to Outside VLAN

```
interface ethernet0/0
switchport access vlan 2
no shutdown
```

7. ASA Configuration - Inside Interface

```
interface vlan 10
nameif inside
security-level 100
ip address 10.0.0.1 255.255.255.0
no shutdown
```

8. Assign Ethernet0/1 to Inside VLAN

```
interface ethernet0/1
switchport access vlan 10
no shutdown
```

9. Configure NAT

```
object network INSIDE-NET
subnet 10.0.0.0 255.255.255.0
nat (inside,outside) dynamic interface
```

10. Configure Default Route

```
route outside 0.0.0.0 0.0.0.0 192.168.0.1
```

11. Enable ICMP Inspection

```
class-map inspection_default
 match default-inspection-traffic

policy-map global_policy
 class inspection_default
  inspect icmp

service-policy global_policy global
```

12. PC Configuration

```
Desktop → IP Configuration

IP Address      : 10.0.0.5
Subnet Mask     : 255.255.255.0
Default Gateway : 10.0.0.1
```

13. Verification Commands

```
ASA Commands:
show interface ip brief
show route
```

```
show run
show nat

Router Commands:
show ip interface brief
show running-config
```

14. Testing Procedures

```
Test 1:
From PC:
ping 10.0.0.1

Expected Result:
SUCCESS

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Test 2:
From ASA:
ping 192.168.0.1

Expected Result:
SUCCESS

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Test 3:
From PC:
ping 192.168.0.1

Expected Result:
SUCCESS
```

15. Important ASA Concepts

```
1. ASA 5505 uses VLAN interfaces for Layer 3 configuration.
2. Ethernet ports behave like Layer 2 switchports.
3. IP addresses are assigned to VLAN interfaces, not physical ports.
4. NAT is required for inside users to access outside networks.
5. ICMP inspection allows ping replies through ASA.
6. Security Levels:
   - inside = 100
   - outside = 0

Higher to lower traffic is allowed by default.
Lower to higher traffic is blocked by default.
```

16. Troubleshooting Guide

```
Problem: PC cannot ping ASA
Check:
- PC IP address
- Default gateway
- Ethernet0/1 assigned to VLAN10

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Problem: ASA cannot ping router
Check:
- Router interface IP
- Router interface status
- Ethernet0/0 assigned to VLAN2

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Problem: PC cannot ping router
Check:
- NAT configuration
```

```
- Default route
- ICMP inspection

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Problem: Interfaces show DOWN
Check:
- no shutdown
- Correct cables
- Correct VLAN assignment
```

17. Final Working Traffic Flow

```
PC (10.0.0.5)
      ↓
ASA Inside (10.0.0.1)
      ↓ NAT
ASA Outside (192.168.0.3)
      ↓
Router (192.168.0.1)
```

18. Skills Learned

```
- Cisco ASA Firewall Basics
- VLAN Interfaces
- NAT/PAT
- Default Routing
- Firewall Security Levels
- ICMP Inspection
- Enterprise Network Design
- Packet Tracer Troubleshooting
- Firewall Traffic Flow
```